

Isaac Watts

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Professional Summary

I hold a Bachelor of Science in Computer Science and Mathematics (dual major) from Virginia State University, graduating summa cum laude. With a strong interest in data science, I have gained experience in machine learning and robotics through academic research projects and past internships. These experiences have developed my ability to work with complex technical systems and derive insights from structured and unstructured data. Outside of academics, I enjoy activities that reinforce logical thinking and pattern recognition, such as chess, puzzles, solitaire, and strategic gaming, which complement my analytical mindset.

Technical Skills

Programming Languages: Python, C, C++, Java, C#, Kotlin, MATLAB, LaTeX

Databases: Relational databases (MySQL, PostgreSQL, and SQLite) and Nonrelational databases (MongoDB)

Data Processing & BI Tools: Pandas, NumPy, Matplotlib, Tableau, Excel

Version Control & Containerization: Git, GitLab, Docker

Operating Systems: Windows 11 and Ubuntu 22.04

Experience

Robotics Student Researcher

Adelphi, MD

Army Research Lab

May 2024 – August 2024

- Trained an AI agent using Unity observations to re-establish radio communications through reinforcement learning by simulating radio waves with C# via Unity Machine Learning Agents Toolkit (ML-agents) and the library PyTorch.

Tutor

Petersburg, VA

Virginia State University

September 2023 – May 2024

- Tutored peers in SQL, R, Python, and data-focused subjects such as discrete math, calculus, and statistics.

Robotics Student Researcher

Adelphi, MD

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- Developed an image processing tool to convert an 8-bit image to a 16-bit format. Compressed that 16-bit image using an image transport layer that allowed the data to be passed through the network using C++ (ROSCPP).
- Automated writing Robot Visualization files for multi-robot experiments by using a Python (ROSPY) automation script, resulting in no longer needing to manually do everything in an RVIZ GUI or file, boosting productivity.

Education

Virginia State University

GPA: 3.9

Bachelor's Degree in Computer Science

- **Coursework:** Intro to Data Science, Probability & Statistics for CS, and Digital Image Processing

Virginia State University

GPA: 3.9

Bachelor's Degree in Mathematics

- **Coursework:** Linear Algebra and Discrete Math

John Tyler Community College

GPA: 4.0

Associates in General Studies

Projects

Data Visualizations from CDC dataset

github.com/IwattsX/ 

- Cleaned data and filtered out null values directly within Tableau to ensure accurate visualization.
- Built a multi-page dashboard exploring trends in chronic diseases by state (mainly Virginia), gender, and demographics.
- Leveraged Tableau's storytelling tools to highlight key insights on health disparities and risk factors.
- Designed interactive elements (filters, highlighters, parameter controls) for user-driven exploration.
- https://public.tableau.com/app/profile/isaac.watts/viz/Disease_indicators_and_insights_IW/Story1 

Inverse Problem Modeling and ML

github.com/IwattsX/med_bodies 

- Simulated voltage readings from a circular mesh with an embedded inclusion using MATLAB.
- Trained a neural network to localize the inclusion based on voltage patterns using Python (ELU + MSE loss + Adam optimizer) using PyTorch.
- Reconstructed 2D images of the mesh to visualize inclusion location from the neural net's predictions.
- Combined physical modeling and machine learning to solve an inverse imaging problem.

Steam Scouts

github.com/IwattsX/Digi-Dynamics 

- Created a MySQL-backed Django web app to collect and explore Steam game metadata from Steam's web API.
- Parsed and stored structured data (e.g., game type, price, DLC) using Python and SQL connector.
- Enabled filtering and querying by game type, release date, and content type.

Robotic Arm Simulation using Isaac Sim

github.com/stars/IwattsX/lists/robotsim 

- Created a ROS2 node using rclpy (Python) that publishes Joint State data using Omniverse's Isaac Sim and Isaac Lab, resulting in a ROS2 node that consistently publishes the correct Joint State data of the robot in the simulation.
- Graphed hand data in Matplotlib that was generated from an Oculus Quest alongside Unity's game engine.
- Created and modified URDF files of a robot hand to visualize and control the joints.

Certificates

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| ◦ Anaconda Python for Data Science Professional Certificate | May 2025 |
| ◦ Learning Python (2021) | May 2025 |
| ◦ Statistics Foundations 1: The Basics | May 2025 |
| ◦ Statistics Foundations 3: Using Data Sets | May 2025 |
| ◦ Introduction to Data Science | May 2025 |
| ◦ Learning Excel: Data Analysis | May 2025 |
| ◦ MATLAB Programming Techniques | March 2025 |
| ◦ MATLAB Fundamentals | February 2025 |
| ◦ Intermediate Android Development Course for Virginia State University | December 2024 |

Publications

A Transformer Approach for Camera-to-LIDAR Data Registration

October 2024

Ju Wang, Yong Tang, Venkat R. Dasari, Billy Geerhart, Brian Rapp, Peng Wang, Wei-Bang Chen, **Isaac Watts**,

[10.1109/IRI62200.2024.00072](https://doi.org/10.1109/IRI62200.2024.00072) 